

Villanova University

Department of Mechanical Engineering

ME 4850 SPRING 2026

**INTERIM REPORT 2. THERMAL FLUID DESIGN OF A SERVER
LIQUID COOLING SYSTEM**

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1. Introduction

Processor chips are high-powered computing machines that require immense cooling systems to maintain a low temperature to ensure the chips can operate at maximum power and efficiency. This project was centered around the design, modeling, optimization, and delivery of a direct-to-chip cooling system for eight processing chips for a next-generation performance server. The first part was completed in this interim report, which detailed the design requirements, established a preliminary model for a straight parallel-fin cold plate to predict the thermal resistance and pressure drop as functions of geometry, TIM resistance, and flow rate. The model was then validated against the experimental cold-plate data provided before finalizing the design for the cold plate.

2. Objectives and Constraints

The objective of the interim report was to design a preliminary cold plate system, made of copper, to effectively cool the processor chip without unnecessary pressure losses. Mitigating the pressure loss was necessary for reducing the pump power that was used to push water through the system. The given constraints for the preliminary cold plate design are shown in Table 1.

Table 1. Cold Plate Constraints

Parameter	Value
Cold Plate Material	Copper
Inlet Water Temperature	20°C
Maximum Chip Temperature	60°C
Maximum Chip Power	2 kW
Chip Size	50x50 mm

The cold plate system consisted of three elements: the fin array, the copper base, and a thermal interface material (TIM). The fin array was connected to the chip using a TIM. The heat generated from the chip conducted through the TIM, conducted through the base of the fin array, and was transferred by convection to the water flowing through the array. A diagram of the system is shown below in Figure 1.

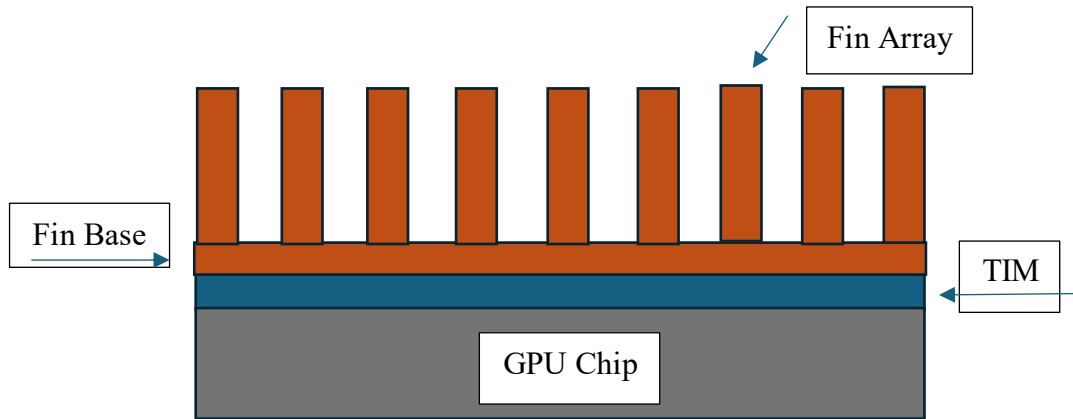


Figure 1. Cold Plate System

3. Development of Cold Plate Model

The initial model was developed in Excel to simulate a cold plate with rectangular fins (see Appendix A). The goal was to use parameters such as the geometry of the fin array and incoming flow rate and produce a total thermal resistance value for the system as well as an estimated pressure drop. The targeted design thermal resistance value was calculated by modeling the cold plate system as three resistors in series, as shown in Figure 2.

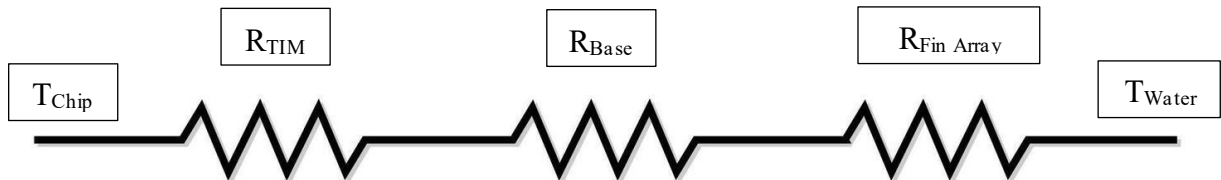


Figure 2. Cold Plate System Resistors

The target design resistance was the sum of the three resistors shown above. It was approximated using the thermal version of Ohm's Law, where the temperature difference (ΔT) between the chip surface and the water temperature (total temperature difference across the resistors) is divided by the total power produced by the chip (Q), which was a given input.

$$R_{System} = \frac{\Delta T}{Q} \quad (1)$$

The values for ΔT and Q were given parameters, so the target thermal resistance for the system was constant throughout the parametric design of the cold plate and had a value of 0.02 °C/W. The chosen design needed to be close to the target resistance to ensure effective cooling and minimize unnecessary pressure loss.

The first step after determining the target resistance was to analyze the flow of the water through the fins. The first calculation was the Reynolds number (Re), which divided the product of the velocity of the water (U_W) and the hydraulic diameter of the channel (D_H) by the kinematic viscosity of the water (ν) at the inlet temperature of 20°C. This calculation showed that the flow through the fins was laminar, since the Reynolds number was consistently below 200, far below the threshold for transition to turbulent flow. The hydraulic diameter was as a function of the cross-sectional area of the channel (A_C) and the perimeter of the channel (P_c). The velocity of the water was calculated as a result of the total volumetric flow rate and the cross-sectional area of the total number of fins (N).

$$Re = \frac{U_W * D_H}{\nu} \quad (2)$$

$$D_H = \frac{4 * A_{C \text{ channel}}}{P} \quad (3)$$

$$U_W = \frac{V}{A_{C \text{ channel}} * N} \quad (4)$$

Then the convective heat transfer coefficient (h) was calculated based on the flow characteristics determined above. The equation for h divided the product of the Nusselt correlation (Nu) and the thermal conductivity of the water (k_l) by the hydraulic diameter.

$$h = \frac{Nu * k_l}{D_H} \quad (5)$$

The Nusselt number was determined by evaluating the effect the entrance region (L_e) had on the total flow. The length of the entrance region was approximated using Equation 6, considering the Reynolds number and the hydraulic diameter.

$$L_e = 0.06 * Re * D_H \quad (6)$$

Even at maximum flow rates, the entrance region was no more than 10% of the total channel length (see Appendix B, Fig. B-1). It was assumed that this had minimal effects on the overall Nusselt number and was deemed negligible to the average convective coefficient for the channel. The Nusselt number was assumed to be 7.54, which is the asymptotic value for fully developed laminar flow in a rectangular channel [1].

With the convective heat coefficient calculated, the fin array efficiency for the input geometry could be calculated. Equation 7 calculated the efficiency of a singular fin, which was defined as the ratio between the heat transfer rate of the fin (q_f) and the maximum heat transfer rate (q_{max}),

which would occur if the entire fin were isothermal at the base temperature. The heat transfer rate of the fin was calculated by assuming an adiabatic fin tip condition.

$$\eta_f = \frac{q_f}{q_{max}} \quad (7)$$

Equation 8 calculated the variable m by taking the square root of the product of the perimeter of the fin cross-section (P_f) and the convective heat coefficient (h) divided by the product of the cross-sectional area of the fin ($A_{c\ fin}$) and the thermal conductivity of the fin material (k_c), which was given as copper. The fin heat transfer rate was determined by Equation 9, where the hyperbolic tangent of the product of L (total length of the fins) and the variable m (calculated with Equation 8) is divided by the product of L and m .

$$m = \sqrt{\frac{h * P_f}{k * A_{c\ fin}}} \quad (8)$$

$$q_f = \frac{\tanh(m * L)}{m * L} \quad (9)$$

The maximum heat transfer rate of the fin (q_{max}) was calculated using Equation 10, which found the product of θ_b (the temperature difference between the base of the fins and the water temperature), A_f (the total exposed surface area of one fin), and h .

$$q_{max} = h * A_f * \theta_b \quad (10)$$

With the efficiency of a singular fin calculated, the total fin array efficiency (η_0) was found using Equation 11. The total fin array efficiency (η_0) was calculated based on the ratio of the fin surface area (A_f) times the number of fins (N) and the entire fin array, then related to the efficiency of a single fin.

$$\eta_0 = 1 - \frac{N * A_f}{A_t} (1 - \eta_f) \quad (11)$$

With the total fin array efficiency calculated, the resistance of the fin array ($R_{Fin\ Array}$) was calculated. The equation for the resistance of a fin array accounts for the product of the mass flow rate ($\dot{m}$) and specific heat (c_p) of the water exchanging heat with the fin array, which is the thermal capacity rate, and the overall fin efficiency rate (η_0), the convective coefficient (h), and the total area of the fin array (A_t).

$$R_{fin\ array} = \frac{1}{\dot{m} * c_p (1 - \exp\left(\frac{-\eta_0 * h * A_t}{\dot{m} * c_p}\right))} \quad (12)$$

The resistance of the fin array was one of three resistance values needed to find the total resistance of the cold plate system. The resistance of the base of the cold plate was a conduction resistance equation. It divided the thickness of the base (t_b) by the product of the thermal conductivity of the copper (k_c) and the total area (A_b) of the base. The calculation for the

resistance of the thermal interface material (R_{TIM}) was the same, except that the conductivity and thickness were different values.

$$R_{Base} = \frac{t}{k_c * A_b} \quad (13)$$

With the Reynolds number determined, Equation 14 calculated the fully developed friction factor (f_F) by dividing a polynomial function including multiple ordered terms with experimentally determined coefficients of the aspect ratio (b/H) by the Reynolds number (Re).

$$f_F = \frac{24 - 32.527 \left(\frac{b}{H}\right) + 46.721 \left(\frac{b}{H}\right)^2 - 40.829 \left(\frac{b}{H}\right)^3 + 22.954 \left(\frac{b}{H}\right)^4 - 6.089 \left(\frac{b}{H}\right)^5}{Re} \quad (14)$$

Equation 15 calculated the dimensionless length parameter (L^+) by dividing the ratio of the channel length (L) to hydraulic diameter (D_h) by the Reynolds number (Re).

$$L^+ = \frac{L/D_h}{Re} \quad (15)$$

The apparent friction factor ($f_{F,app}$) was calculated in Equation 16 by dividing the square root of the sum of two terms by the Reynolds number. The first term was the quantity 3.44 divided by the square root of the dimensionless length parameter (L^+) squared. The second term was the product of the fully developed friction factor (f_F) and the Reynolds number (Re), squared.

$$f_{F,app} = \frac{\sqrt{\left(\frac{3.44}{\sqrt{L^+}}\right)^2 + (f_F * Re)^2}}{Re} \quad (16)$$

The pressure drop across the channel due to major head loss (ΔP_{Major}) was calculated using Equation (17). ΔP_{Major} was equal to the product of four times the apparent friction factor ($f_{F,app}$) multiplied by the ratio of the channel length (L) to the hydraulic diameter (D_h) and the dynamic pressure term, defined as half of the product of the fluid density (ρ) and the square of the fin channel velocity (U_W).

$$\Delta P_{Major} = (4f_{F,app}) \frac{L}{D_h} \frac{\rho U_W^2}{2} \quad (17)$$

Equation 18 calculated the porosity (σ). Porosity was the ratio of the fin array's open cross-sectional area (A_{open}) to total cross-sectional area (A_T). This was calculated by dividing the product of the quantity of the fin array's total width (W) minus the number of fins (N) and the fin height (H) by the product of H and W .

$$\sigma = \frac{(W - N) * H}{W * H} \quad (18)$$

Equation 19 calculated the entrance loss coefficient (K_C). K_C was calculated by multiplying the quantity of 1 minus sigma (σ) squared by 0.42. The negative exit loss coefficient ($-K_E$) was also calculated using Equation 19, with a coefficient of -1 rather than 0.42.

$$K_C = 0.42(1 - \sigma^2) \quad (19)$$

Equation 20 calculated the pressure drop at the channel entrance due to minor head loss (ΔP_C). ΔP_C was equal to the product of the entrance loss coefficient (K_C) and the dynamic pressure term, defined as half of the product of the fluid density (ρ) and the fin channel velocity (U_W) squared. The pressure gain at the channel exit due to minor head gain (ΔP_E) was also calculated using Equation 18, but with the use of the negative exit loss coefficient ($-K_E$).

$$\Delta P_C = K_C \frac{\rho U_W^2}{2} \quad (20)$$

Equation 21 calculated the thermal capacity rate (C_{min}) by multiplying the specific heat of water (c_p) and the mass flow rate of water ($\dot{m}$).

$$C_{min} = \dot{m} * c_p \quad (21)$$

Equation 22 calculated the design's optimal NTU by finding 1 over the product of the total thermal resistance (R_T) and the thermal capacity rate (C_{min}). This equation simplification was made by incorporating the fact that $NTU = UA/C_{min}$ and $UA = 1/R_T$.

$$NTU = \frac{1}{C_{min} * R_T} \quad (22)$$

4. Model Validation

The model demonstrated above was validated by comparison to experimental data provided. The overall fin array resistance and the overall pressure drop were calculated and compared for the provided cold plate geometry, flow specifications, and temperatures. The comparisons are shown below in Figures 3 and 4. While the resistance values were almost identical, the pressure drop values differed. This difference was similar between each data point, but this discrepancy will continue to be examined for the final report.

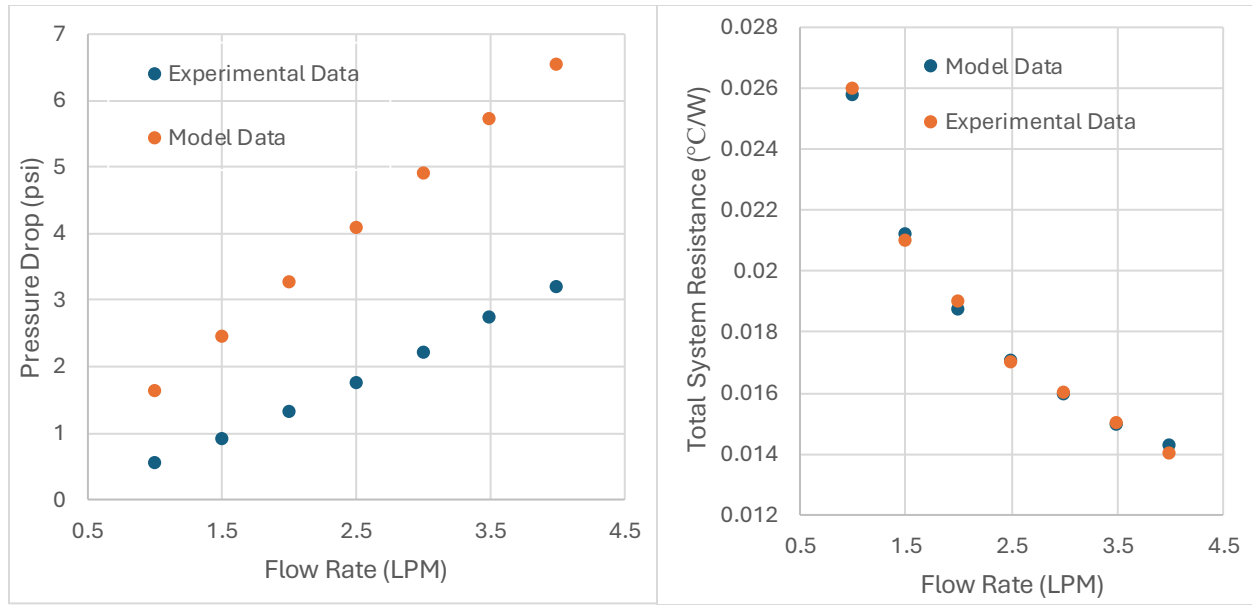


Figure 3. Pressure Drop vs Flow Rate Comparison and Cold Plate System Resistance vs Flow Rate Comparisons for Modeled Data vs Experimental Data

5. Selected Cold Plate Design

The three major factors considered for the preliminary design selection were the pressure drop through the fin array, the total system resistance, and the fin array efficiency. The goal with the total thermal resistance was to get as close to the ideal resistance as possible without exceeding it, to ensure the cold plate effectively cooled the chip below the maximum temperature of 60°C without unnecessary additional pressure loss. The resistance was also manipulated by changing other factors, such as the base thickness, the TIM thickness, and the material of the TIM. Since this value could be adjusted after picking the fin array design, the cold plate itself was analyzed for pressure drop and efficiency. For selecting the number of fins, the overall fin array resistance and the pressure drop were analyzed, as shown below in Figures 3 and 4.

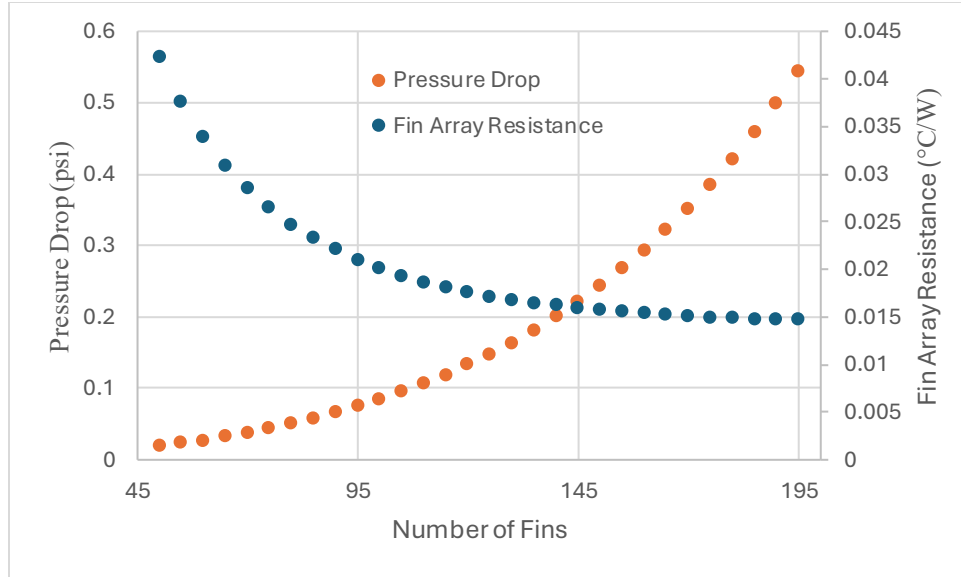


Figure 4. Pressure Drop and Fin Resistance vs Number of Fins at 1 LPM

The target thermal resistance was calculated to be 0.02 °C/W for the system, shown in Figure 3. The thermal resistance of the fin array did not go below 0.02 until over 105 fins were in the array. The pressure drop through the array increased as the fin count increased. Based on this correlation, the fin count should also be minimized. The final design was set at 120 fins, the lowest acceptable count at which the fin array efficiency was below the target thermal resistance for the system.

The height of the fins had major effects on both the fin array efficiency and the pressure drop. The taller the fins, the more the temperature decreased towards the tip of the fin, which lowers the fin efficiency. At constant flow rates, the increased height also lowered the pressure drop through the fins by increasing the hydraulic diameter, which is inversely proportional to the major head loss as shown in Equation 17. The relationship between the height of the fins and the pressure drop and efficiency was represented in the graphs in Figure 5. The fin height was evaluated between 1 and 7 mm.

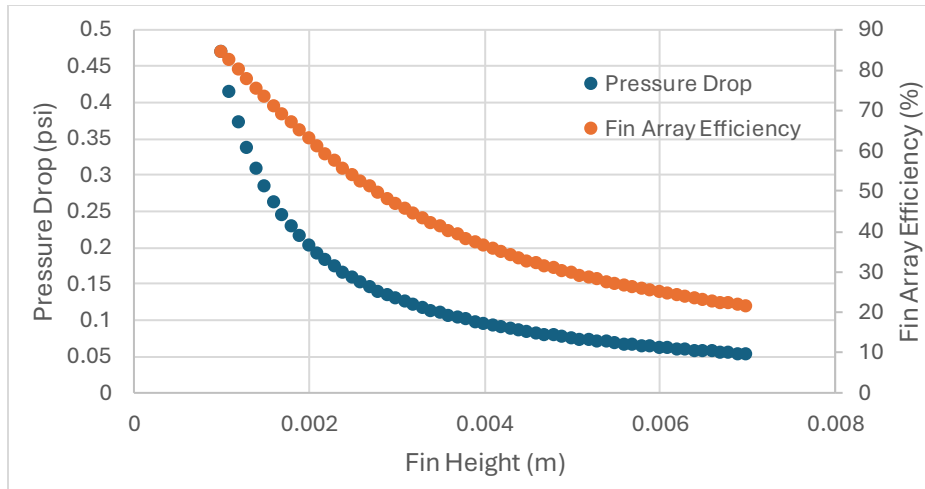


Figure 5. Pressure Drop and Fin Array Efficiency vs Fin Height for 1.5 LPM

The fin array efficiency dictated how much material was unnecessary for the convection of heat to the cooling liquid. The loss of pressure was a recurring operating cost, while the lower fin efficiency was a production cost. The chosen height was 3mm, resulting in an array efficiency of 50.1%. This height maintained a cheaper manufacturing process while still keeping pressure drop to .13 psi.

The flow rate of the cooling liquid was the most important factor in determining the cooling capacity, overall resistance, and pressure drop of the system, which is why it was analyzed after an optimal geometry was chosen. In the fin array resistance calculation shown in Equation 12, the mass flow rate had a major effect on the total resistance value. Additionally, the amount of water moving through the system had noticeable effects on the pressure drop. The effects of the flow rate on the resistance and pressure drop are shown below in Figure 6.

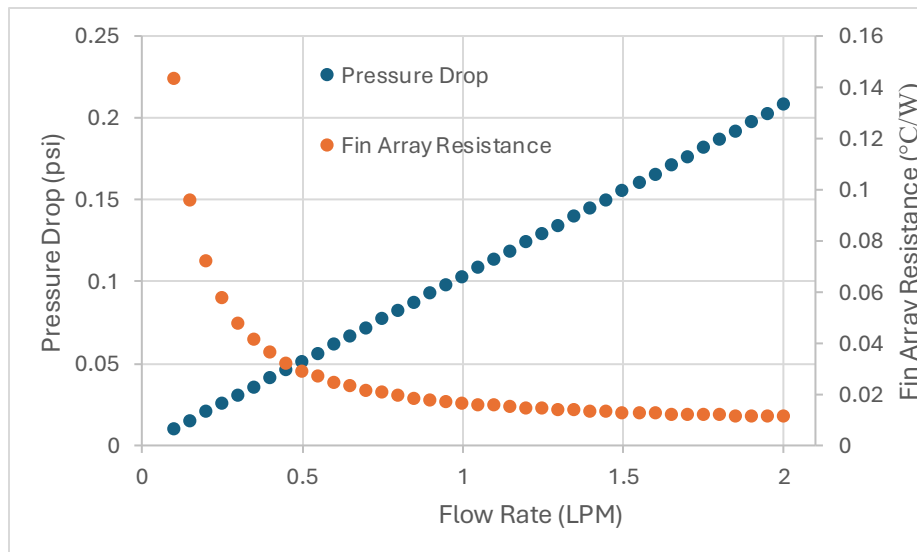


Figure 6. Pressure Drop and Fin Array Efficiency vs Flow Rate

The determination of the minimum flow rate was the same as the number of fins, that the array resistance must be below the target system resistance of $0.02\text{ }^{\circ}\text{C/W}$. Figure 6 shows that the resistance of the array drops below 0.02 at 0.8 LPM , which became the minimum flow rate. The pressure drop increased proportionally to the flow rate, so the optimal flow rate would also be minimized as much as possible. The flow rate would also have to result in a cold plate resistance at such a value that the TIM resistance was realistic. The thermal conductivity values for high-end TIMs was $5\text{-}15\text{ W/mK}$ [2]. So, the flow rate would have had to result in a necessary k value in that range, which was calculated using the difference between the ideal system resistance and the fin array resistance plus the base resistance to get the ideal TIM resistance.

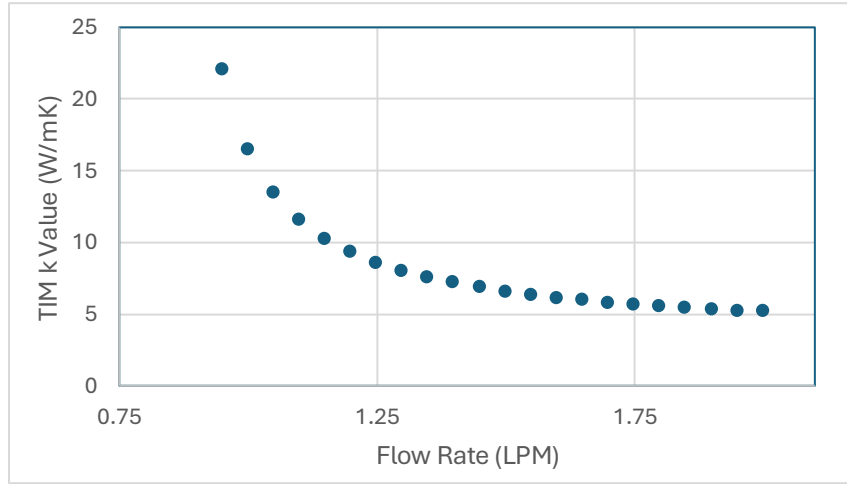


Figure 7. TIM Thermal Conductivity vs Flow Rate

At 1.2 LPM , the fin array resistance is below the ideal system resistance, with enough difference to account for the base and TIM resistance. It also minimizes the pressure drop, and results in a TIM resistance and thermal conductivity within the range for effective TIM materials. Some options for the TIM would be ARCTIC MX-4 or Thermal Grizzly TG-A-014-R, two likely options for a thermal interface grease, which are both around 8.5 W/mK . Additionally, there were options for thermal interface pads, such as the Thermal Grizzly Kyrosheet or the Thermo Grizzly Phasesheet, also both at 8.5 W/mK . The thickness was selected to be 1.0 mm , a middle point for the typical thicknesses of greases and thermal pads [2].

6. Cold Plate Design Specifications

Based on the above analysis and considerations, the preliminary design for the cold plate for a singular processing chip was established, as presented below in Table 1. A CAD model for the cold plate is also shown below in Figure 8.

Table 2. Preliminary Cold Plate Design Specifications

Parameter	Value
Total Thermal Resistance	$.02\text{ }^{\circ}\text{C/W}$

TIM Resistance	0.00428 °C/W
Cold Plate Thermal Resistance	0.01468 °C/W
TIM Thickness	0.1 mm
TIM Thermal Conductivity	~9.34 W/mK
Fin Height	3 mm
Fin Thickness	0.1 mm
Fin Spacing	0.32 mm
Fin Length	50 mm
Base Thickness	1 mm
Base Resistance	0.001309 °C/W
Fin Array Efficiency	50.1%
Number of Fins	120
Shape of Fins	Parallel Rectangular
NTU	0.598
Volumetric Flow Rate	1.2 LPM
Total Pressure Drop	0.124 PSI

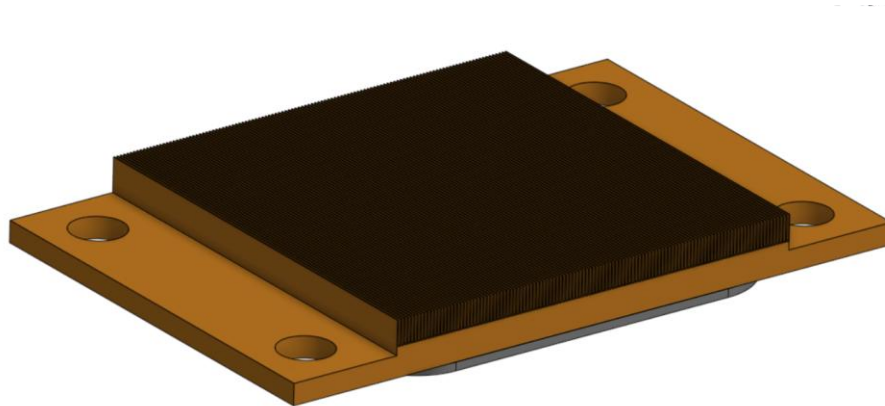


Figure 8. CAD Model of Preliminary Cold Plate System Design

References

- [1]Lienhard, J. H., 2025, “On the Nusselt Number for Thermally Developed Flow Between Isothermal Parallel Plates With Dissipation,” ASME Digital Collection [Online]. Available: <https://asmedigitalcollection.asme.org/heattransfer/article/147/11/111801/1219113/On-the-Nusselt-Number-for-Thermally-Developed-Flow>. [Accessed: 08-Feb-2026].
- [2]Jiang, S., 2025, “Thermal interface material (Tim): Selection guide,” Voltera Inc. [Online]. Available: <https://www.voltera.io/blog/thermal-interface-material-tim?srsltid=AfmBOoozNLT2ZiFHFayNPY682yz1RbcV9Z6YIwirOse3-GrwmxcqX14s>. [Accessed: 10-Feb-2026].

Appendix

A. Excel Spreadsheet Model for Parallel Straight-Finned Cold Plate

[ColdPlate model.xlsx](#)

B. Developing Flow Entrance Region Length vs Flow Rate for Preliminary Design

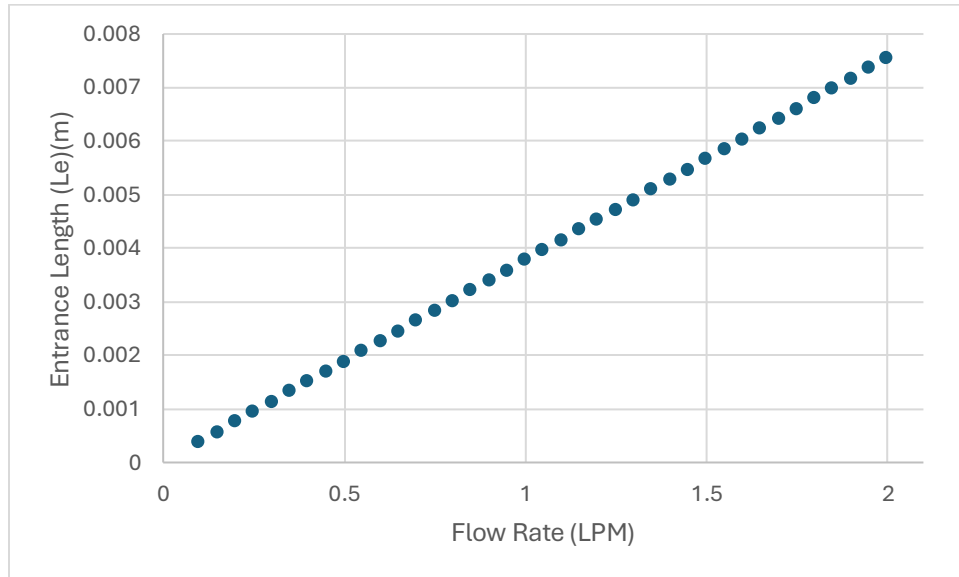


Figure B-1. Entrance Length vs Flow Rate for Preliminary Design